

What is AI?

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It is a branch of Computer Science that pursues creating the computers or machines as intelligent as human beings.

It is the science and engineering of making intelligent machines, especially intelligent computer programs.

It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable

What is AI? (contd ..)

Definition: Artificial Intelligence is the study of how to make computers do things, which, at the moment, people do better.

According to the father of Artificial Intelligence, John McCarthy, it is “The science and engineering of making intelligent machines, especially intelligent computer programs”.

Artificial Intelligence is a way of making a computer, a computer-controlled robot, or a software think intelligently, in the similar manner the intelligent humans think.

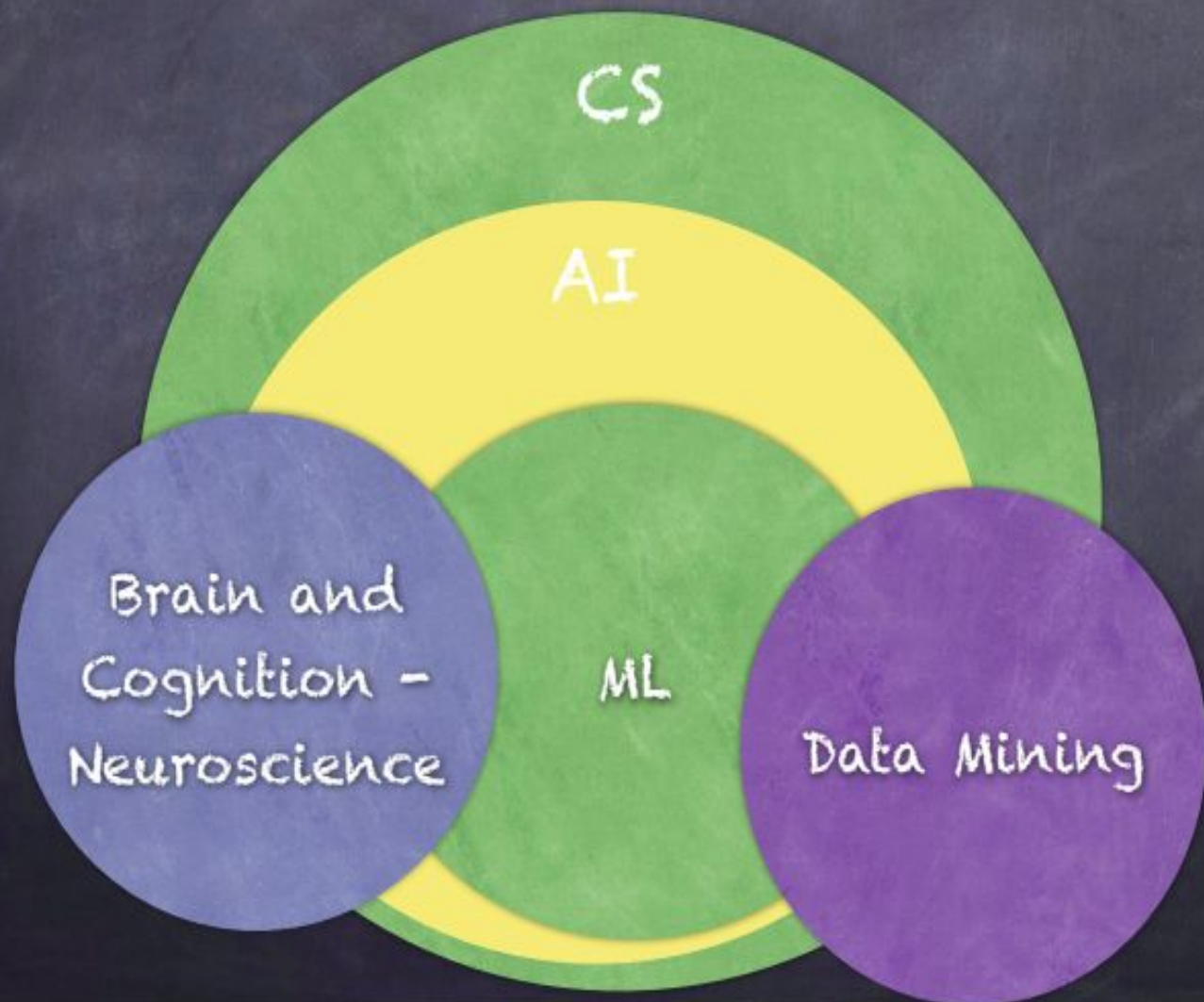
AI is accomplished by studying how human brain thinks and how humans learn, decide, and work while trying to solve a problem, and then using the outcomes of this study as a basis of developing intelligent software and systems.

What is AI? (contd ..)

It has gained prominence recently due, in part, to big data, or the increase in speed, size and variety of data businesses are now collecting. AI can perform tasks such as identifying patterns in the data more efficiently than humans, enabling businesses to gain more insight out of their data.

From a business perspective AI is a set of very powerful tools, and methodologies for using those tools to solve business problems.

From a programming perspective, AI includes the study of symbolic programming, problem solving, and search.



Applications of AI

- Problem Solving
- Search and Control strategies
- Speech Recognition
- Natural Language Processing
- Computer Vision
- Expert Systems

Problems in AI

Intelligence does not imply perfect understanding; every intelligent being has limited perception, memory and computation. Many points on the spectrum of intelligence versus cost are viable, from insects to humans.

AI seeks to understand the computations required from intelligent behaviour and to produce computer systems that exhibit intelligence.

Aspects of intelligence studied by AI include perception, communication using human languages, reasoning, planning, learning and memory.

Problems in AI (contd..)

The following questions are to be considered before we can step forward:

1. What are the underlying assumptions about intelligence?
2. What kinds of techniques will be useful for solving AI problems?
3. At what level human intelligence can be modelled?
4. When will it be realized when an intelligent program has been built?

AI Definition

We call programs intelligent if they exhibit behaviors that would be regarded intelligent if they were exhibited by human beings.

-- Herbert Simon

AI is the study of techniques for solving exponentially hard problems in polynomial time by exploiting knowledge about the problem domain.

-- Elaine Rich

ELIZAnamed after Eliza Doolittle, a working-class character in George Bernard Shaw's play *Pygmalion*,

ELIZA was a simple natural language processing program written at MIT by Joseph Weizenbaum around 1966.

It had simple rules to manipulate language and would essentially modify the users input to generate its response. A popular version called Doctor was running a script designed to make respond like a Rogerian psychotherapist.

A Russian scientist chatting with Doctor at Stanford

Doctor: What brought you here to see me today?

Visitor: Oh, nothing much. I'm feeling a bit tired, that's all.

Doctor: Why do you think you're feeling a bit tired.

Visitor: Well I've been traveling a lot, and away from home.

Doctor: Tell me about your family.

What AI can do?

- **Learning - e.g. Text Categorization**
- **Games - Chess Champion**
- ..

What AI can't do?

- Understand natural language robustly
- Surf web
- Interpret an arbitrary visual scene
- Learn a natural language.

AI Techniques

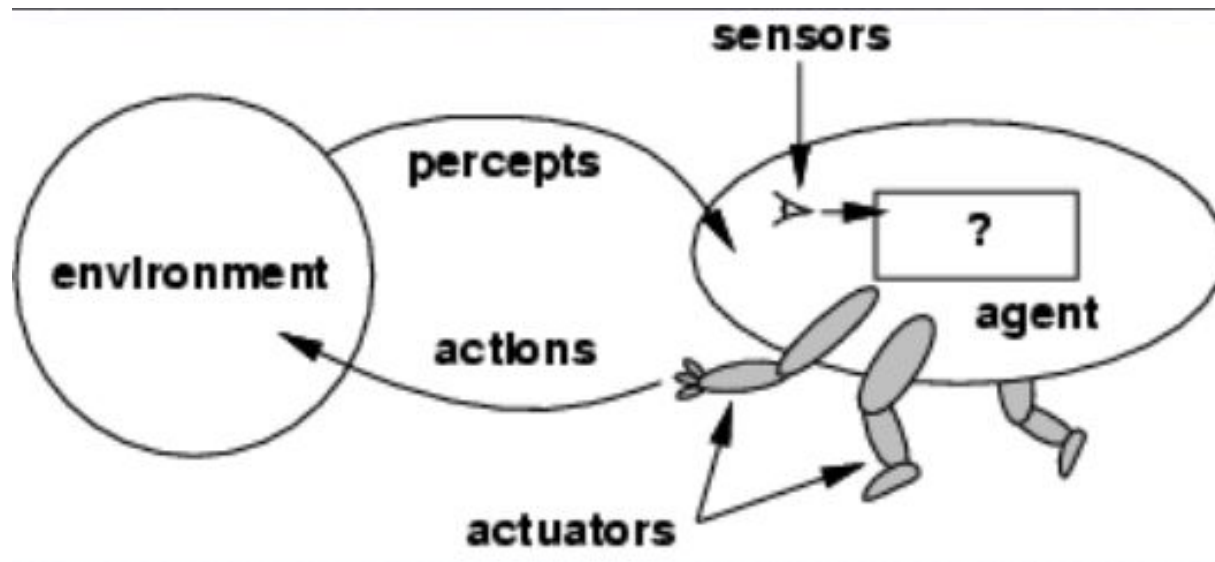
- Search
- Knowledge
- Abstraction

⑥ Fundamental technique

⑥ Possible answers, decisions or courses of action are structured into an abstract (hypothesis) space, which we then searched.

Intelligent Agent

- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.



Types of Intelligent Agent

- **Human Agent:** Eyes, Ears and other organs for sensors; hands, legs, mouth and other body parts for actuators.
- **Robotic Agent:** Cameras and Infrared range finders for sensors; various motors for actuators

Rational Agents

An agent should strive to “do the right thing”, based on what it can perceive and the actions it can perform. The right action is the one that will cause the agent to be most successful.

Performance Measure: An object criterion for success of an agent's behavior.

E.g. performance measure of a vacuum cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated etc.

Rational Agents (contd..)

For each possible percept sequence, **a rational agent** should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Autonomous Agent

- A system is autonomous to the extent that its own behavior is determined by its own experience.
- Therefore, a system is not autonomous if it is guided by its designer according to a priori decisions.
- To survive, agents must have:
 - Enough built-in knowledge to survive
 - The ability to learn

PEAS

- PEAS:
 - Performance Measures
 - used to evaluate how well an agent solves the task at hand
 - Environment
 - surroundings beyond the control of the agent
 - Actuators
 - determine the actions the agent can perform
 - Sensors
 - provide information about the current state of the environment

PAGE

- PAGE:
 - Percepts
 - information acquired through the agent's sensory system
 - Actions
 - operations performed by the agent on the environment through its actuators
 - Goals
 - desired outcome of the task with a measurable performance
 - Environment
 - surroundings beyond the control of the agent

StudentBot PEAS Description

- Performance Measures grade
 time spent studying
 career success
- Environment classroom, university, universe
- Actuators human actuators
- Sensors human sensors

StudentBot PAGE Description

- Percepts images (text, pictures, instructor, classmates)
sound (language)
- Actions comments, questions, gestures
note-taking (?)
- Goals mastery of the material
performance measure: grade
- Environment classroom

Example: Automatic Taxi Driving System

Percepts: Video, sonar, speedometer, odometer, engine sensors, keyboard input, microphone, GPS, ...

Actions: Steer, accelerate, brake, horn, speak/display, ...

Goals: Maintain safety, reach destination, maximize profits (fuel, tire wear), obey laws, provide passenger comfort, ...

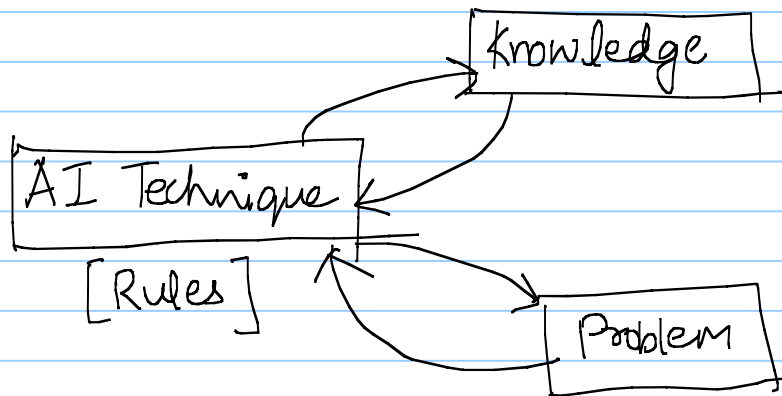
Environment: U.S. urban streets, freeways, traffic, pedestrians, weather, customers, ...

Different aspects of driving may require different types of agent programs!

AI (Lec-4)

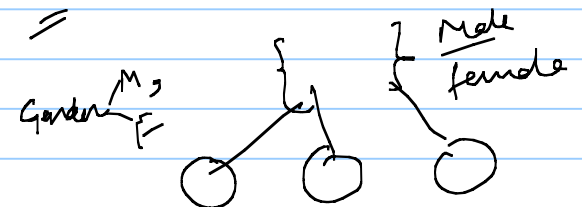
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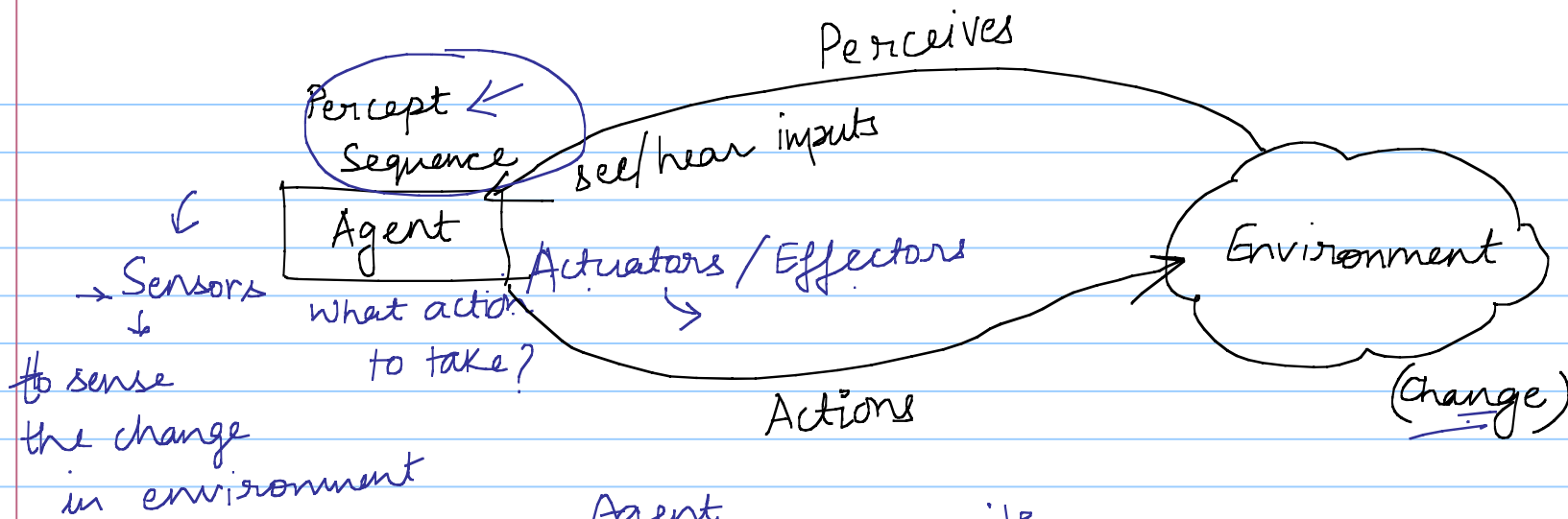
13-01-2021



Knowledge

- ⇒ large in volume
- ⇒ dynamic (constantly changing)
- ⇒ not well formatted





Agent

↳ operates in its environment.

→ perceives its environment through sensors.

→ acts upon environment through actuators / effectors.

① Agent perceives its environment (Role of Sensors)

→ Complete set of input at a given time is called percept.

i/p
percept
seq.

→ current percept, or percept sequence can influence the actions of agent.

② Agent can change the environment (through actuators/ effectors)

o/p
action seq.

→ An operation involving an actuator is called Action.

→ Actions can be grouped into Action sequences.

Agent program

→ implements mapping from percept sequence to action sequence.

Agent $\leftrightarrow$ Intelligent Agent

How to evaluate performance of Agent ?

- ↳ Behavior and performance of agent in terms of agent funcⁿ.
 - ↳ " Perception sequence history to action mapping.
 - ↳ " Ideal mapping : specifies ^{actions} which agent should take

↳ Performance Measure

- " a subjective measure which characterize how successful a agent is.
- e.g. Speed, Power usage, Accuracy etc.

Example of Agent

- Humans
 - Eye, Ear etc. (sensors) ←
 - hands, legs etc. (actuators) ←
 - Agent pgm. (brain).
- Calculators
- Robots
 - Sensors: Thermal, solar, infrared, camera etc.
 - Actuators: Motors, wheels, grippers, lights, speakers etc.
- Softbots (Software Agents)
 - Sensors: funcⁿ (reading)
 - Actuators: funcⁿ (writing)

- Expert Systems in healthcare : Cardiologists.
- Autonomous Spacecraft : Mars Rover
- Intelligent Buildings : → Lighting condⁿ, Air conditioning, Doors, Fire sensors.

Basic Functionalities of Agent :-

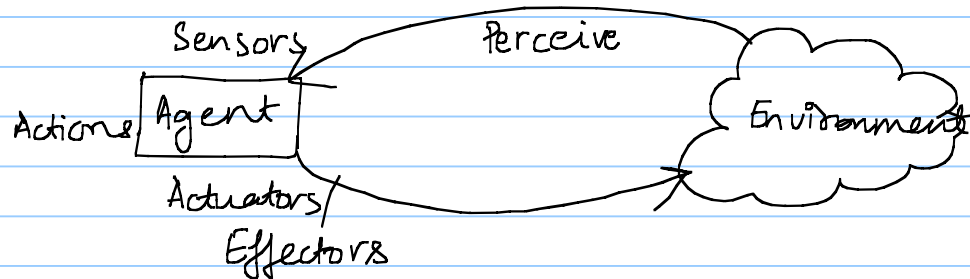
- 1. Sensing (must sense)
- 2. Acting (must act) pre-requisite for autonomous agents.
- 3. Understanding, Reasoning, Learning

18/1/2021

Note Title

Artificial Intelligence

17-01-2021



Rational Agent

→ which always do the right thing

In reality, a Rational Agent is not omniscient

→ does not know actual outcome of its actions.

→ it may not know certain aspects of its environment.

Rationality must take into account the limitations of the agent.

→ percept sequence, feasible actions, background knowledge

Rationality

→ Perfect Rationality

→ Bounded Rationality :- property of an agent that behaves in a manner which is nearly optimal with respect to its goals as its resources will allow.

Agent Environment

→ in terms of what it appears to the agent rather than from absolute point of view.
how

① on the basis of Observability of environment.

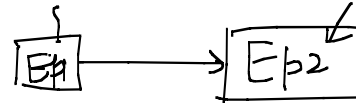
→ Fully observable :- All the environment relevant to the actions being considered is observable. e.g. Chess

→ Partially observable :- Relevant features of environment are partially observable. e.g. Poker

② On the basis of Determinism

- Deterministic :- Next state of environment is completely described by current state and action of agent.
e.g. Image Analysis.
- Stochastic :- If an element or interference or uncertainty occurs, then environment is stochastic.
Deterministic + Partially observable = Stochastic
e.g. Ludo. Uncertainty because of o/p of die roll.
- Strategic :- environment state wholly determined by preceding state and actions of multiple agents.
e.g. chess.

③ On the basis of Episodicity



- Episodic environment :- Subsequent episodes do not depend on what actions occurred in previous episodes
- Sequential environment :- Agent engages in a series of connected episodes.

④ On the basis of dynamism.

- Static environment :- does not change while agent is thinking.
- Dynamic environment :- does change while agent is thinking.

⑤ On the basis of continuity

→ Discrete :- number of distinct percepts and actions is limited

→ Continuous :- " " " " " not limited.

⑥ On the basis of # agents

→ Single agent environment :- contains single agent.

→ Multi-agent environment :- contains multiple agent.

Complexity of Agent Environment

① Knowledge Rich :- If environment contains enormous amount of information.

② Input Rich :- If environment contains enormous amount of input that it can send to agent.

- Sensing Strategies
 - Attentional Mechanisms
- } How agents manage rich environment ?

ARTIFICIAL INTELLIGENCE (Lec-6)

Note Title

19-01-2021

Agent Environment

The Various types of Agents:-

① Table-based Agent:-

- i/p from sensors - percept sequence.
- triggers actions through effectors.
- for all possible percept seq., their mappings to actions are predetermined and stored in a table.
- while running, just lookup in the table for the corresponding action to the current percept.

{ Rule-based
Neural networks
Algorithm. ↓

Disadvantages:- ① Enormous amount of space required.

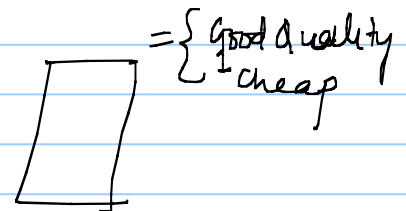
② All actions are predetermined, there is no autonomy.

← Fully observable

② Percept-based Agent :-

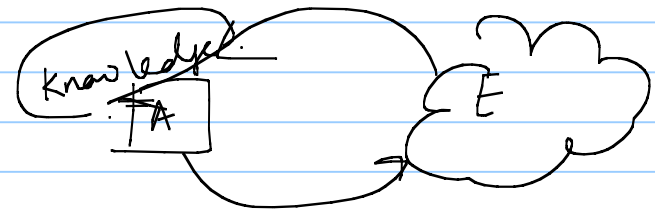
- Simple reflex agent
- based on current i/p (percept). :- There is no history of percept, current state is the only thing it has.
- Reactive Agent, Stimulus Response Agent.

- Efficient
- No strategic planning, learning.
- Not good for multiple opposing goals.



③ State-based Agents :-

- based on State of the world and Knowledge (memory).
- also work for partially observable environment.



④ Goal Based Agent. (Goal(s))

- take decision based on how far they are current from their goal.
- more flexible.
- use Search & Planning.

③ Utility Based Agent

designed based on their end use.

- preference as utility.
 - quick, cheap, safe trip to our destination.
- Utility : "how happy" the agent is.

⑥ Learning Agents

- learn from past experience. , it has learning capabilities
- act and adapt automatically through learning.
- True Autonomy.
- Initially, they work in unknown environments.

Summary

Agent :- An agent perceives and acts in an environment, has an architecture, and is implemented by agent program.

Ideal Agent :- An agent which always chooses the action which maximizes its expected performance.

Autonomous Agent :- uses its own experience rather than built-in knowledge of environment by the designer.

Agent Program :- Maps from percept to action and update its internal state.

Representing Knowledge :-

Environment :- fully observable / partially observable, sequential / episodic
static / dynamic, discrete / continuous, single / multiple-agent.

Rational Agent :- always chooses an action based on percept sequence it has received so as to maximize its expected performance measure given the percepts (i/p) and knowledge preprocessed by it.

Bounded Rationality :- takes best possible decisions given its goal and bounded resources.

Next Topic is

State Space Search

